

Part 4 of 5

American Carrier Strike Groups: An Electronic Perspective



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In the previous parts of this article, we discussed air defence warfare, electronic warfare, under-sea warfare, surface warfare and strike warfare. Let us take a look at air operations in this part.

Air operations

Though air operations also fall under strike warfare, these are specialised operations of a carrier and, in turn, carrier strike group (CSG). Air operation from carriers is highly complex and highly dangerous. To support air operation, a myriad of electronic systems are installed in the carrier.

A combat aircraft is launched from the carrier through a system called catapult, which is driven by steam. Since aircraft on a carrier do not have sufficient distance to run and take off, the catapult proves very useful. An aircraft is placed on shuttle of the catapult and launched; aircraft is airborne in just two seconds. Steam pressure that has to be maintained is controlled through electronic systems much similar to industrial systems.

The catapult launches aircraft in air and

then the aircraft using its full engine power ascends to an altitude. After that, it leaves for its area of responsibility. Ensuing combat actions are out of purview of this article, however the return leg is covered here.

For retrieving and receiving an aircraft on the carrier, a plethora of electronic systems support the eventual landing on the carrier. But before landing, aircraft must navigate towards the carrier.

Navigation

In the sea, even gigantic carriers appear smaller than a dot for their aircraft flying at heights greater than 10km. At those heights, visually sighting the carrier becomes impossible. To retrieve the launched aircraft, the carrier must reveal its position to the incoming aircraft. The carrier does this through a system called TACTical Air Navigation (TACAN).

TACAN. Through a TACAN transponder the carrier continuously relays its identity in Morse code at a particular frequency. Through appropriate antenna arrangement, it maintains a cardioid radiation pattern for this transmission. Aircraft can receive these signals by tuning their TACAN receivers.

These receivers compare aircraft's bearing with the direction of the signal reception and give the bearing (direction) of the carrier to the pilot. The TACAN receiver also gives the slant range, known as crow-flight distance, of the carrier.

For this, aircraft's TACAN receiver interrogates the carrier's TACAN transponder with an interrogate pulse. The TACAN transponder receives that and, after a delay of 50µs, sends a reply signal. Aircraft's TACAN receiver receives this reply. Then, by knowing the time at which the receiver had sent the interrogating pulse and the time it received the reply, it calculates the round-trip distance time. This time minus the 50µs delay is the actual round-trip distance time. With the speed of the radio waves known,

Fig. 23: CATCC; the person sitting at the front is the approach controller

